BRICSCOIN

A Post-Quantum Secure Cryptocurrency

Technical Whitepaper v3.0

Jabo86

February 2026

BricsCoin is a decentralized cryptocurrency built on SHA-256 Proof-of-Work consensus, featuring a hybrid post-quantum cryptographic signature scheme that combines classical ECDSA (secp256k1) with NIST-standardized ML-DSA-65 (FIPS 204). Both signatures are required at the consensus level for transaction validation, providing genuine quantum resistance. BricsCoin is designed for ASIC mining compatibility via the Stratum v1 protocol with BIP320 version rolling support.

Table of Contents

1. Introduction
2. Protocol Architecture
3. Consensus Mechanism
4. Post-Quantum Cryptography
5. Tokenomics
6. Mining Protocol
7. Transaction Model
8. Network Architecture
9. Wallet Ecosystem
10. Security Analysis
11. Conclusion

1. Introduction

The advent of large-scale quantum computers poses an existential threat to the cryptographic foundations of modern blockchain networks. Elliptic Curve Digital Signature Algorithm (ECDSA), the signature scheme used by Bitcoin and the vast majority of cryptocurrencies, is vulnerable to Shor's algorithm, which can efficiently solve the elliptic curve discrete logarithm problem on a sufficiently powerful quantum computer.

BricsCoin addresses this threat proactively by implementing a hybrid signature scheme at the consensus level. Every transaction from a post-quantum (PQC) wallet requires two valid signatures: one from ECDSA (secp256k1) for backward compatibility and one from ML-DSA-65 (formerly Dilithium), a lattice-based digital signature algorithm standardized by NIST in FIPS 204. A transaction is rejected by the network if either signature fails verification.

This is not a superficial or preparatory implementation. BricsCoin enforces hybrid signature verification at the consensus layer, making it one of the first operational blockchains to require post-quantum signatures for transaction validity.

1.1 Design Principles

Quantum Resistance. Hybrid ECDSA + ML-DSA-65 signatures enforced at the consensus level. Both signatures must be valid for a transaction to be accepted by the network.

ASIC Compatibility. SHA-256 Proof-of-Work enables mining with existing Bitcoin ASIC hardware (Bitaxe, Antminer) via Stratum v1 protocol with BIP320 version rolling.

Client-Side Security. All private keys are generated and stored locally. Signing operations occur entirely in the user's browser or desktop wallet. Private keys never leave the device.

Bitcoin-Proven Economics. Fixed supply of 21 million coins, halving every 210,000 blocks, and difficulty adjustment every 2,016 blocks mirror Bitcoin's time-tested economic model.

2. Protocol Architecture

BricsCoin operates as a full-stack blockchain system comprising several interconnected components:

Component	Technology	Purpose
Consensus Engine	FastAPI (Python)	Block validation, transaction processing, chain management
Mining Interface	Stratum v1 Server	ASIC miner communication, job distribution, share validation
Database	MongoDB	Persistent storage for blocks, transactions, wallets, mining stats
Web Frontend	React.js	Block explorer, wallet interface, mining dashboard, network stats
Desktop Wallet	Electron	Cross-platform native wallet with full PQC signing capability
PQC Module	ML-DSA-65 (FIPS 204)	Post-quantum signature generation and verification

Table 1: System Architecture Components

3. Consensus Mechanism

3.1 SHA-256 Proof-of-Work

BricsCoin uses SHA-256 double-hashing for its Proof-of-Work consensus, identical to Bitcoin's mining algorithm. This design choice enables compatibility with the extensive ecosystem of SHA-256 ASIC miners, including consumer-grade devices like the Bitaxe series and industrial miners such as the Antminer S19/S21.

Block validity requires that the SHA-256 hash of the block header, when interpreted as a 256-bit integer, is less than or equal to the current target value:

```
H(block_header) ≤ target

where: target = MAX_TARGET / difficulty
MAX_TARGET = 2^256 - 1
```

3.2 Difficulty Adjustment

The difficulty adjustment mechanism ensures a stable average block time of 10 minutes (600 seconds), regardless of changes in total network hashrate.

Parameter	Value
Target Block Time	600 seconds (10 minutes)

Adjustment Interval	Every 2,016 blocks (~14 days at target rate)
Maximum Adjustment	4x increase or 0.25x decrease per interval
Initial Difficulty	1,000,000
Bootstrap Phase	Every 10 blocks for the first 2,016 blocks

Table 2: Difficulty Adjustment Parameters

The adjustment formula:

```
new_difficulty = current_difficulty x (expected_time / actual_time)

clamped to: 0.25 x current ≤ new_difficulty ≤ 4 x current
```

4. Post-Quantum Cryptography

BricsCoin implements a **Level 1 (strong)** post-quantum security model. The hybrid signature scheme is enforced at the consensus level. Both classical and post-quantum signatures are mandatory for transaction validation.

4.1 Hybrid Signature Scheme

Every PQC transaction carries two independent signatures:

Algorithm	Standard	Key Size	Signature	Security
ECDSA	secp256k1	32 B (priv) / 64 B (pub)	64 bytes	Classical 128-bit
ML-DSA-65	FIPS 204	4,032 B (priv) / 1,952 B (pub)	3,309 bytes	NIST Level 3 (quantum)

Table 3: Hybrid Signature Algorithms

4.2 Consensus-Level Enforcement

The verification function implements strict AND logic:

```
hybrid_valid = ecdsa_valid AND dilithium_valid
```

If **either** signature fails, the transaction is rejected. There is no fallback to single-signature validation. Even if ECDSA is broken by a quantum computer, an attacker cannot forge a valid transaction without also forging the ML-DSA-65 signature, which remains computationally infeasible.

4.3 Address Derivation

PQC wallet addresses are derived from both public keys, cryptographically binding the address to both key pairs:

```
combined_hash = SHA-256(ecdsa_public_key || dilithium_public_key)
address = "BRICSPQ" + combined_hash[0:38]
```

The "BRICSPQ" prefix distinguishes post-quantum addresses from legacy "BRICS" addresses, enabling appropriate signature verification rules.

4.4 Deterministic Key Generation

Both key pairs are generated deterministically from a single 12-word BIP39 mnemonic seed phrase:

```
ecdsa_private_key = seed[0:32]
dilithium_seed = SHA-256(seed[32:64] || 'dilithium-v1')
dilithium_pk, sk = ML_DSA_65.keygen_internal(dilithium_seed)
```

This ensures complete wallet recovery from the seed phrase alone, including both ECDSA and ML-DSA-65 key pairs.

4.5 Client-Side Signing

All cryptographic operations occur locally on the user's device. In the web wallet, JavaScript implementations run in the browser. In the desktop wallet, operations run in the Electron/Node.js process. Private keys are never transmitted to any server. Only the resulting signatures and public keys are sent for verification.

5. Tokenomics

BricsCoin follows Bitcoin's proven deflationary monetary policy:

Parameter	Value
Ticker Symbol	BRICS
Maximum Supply	21,000,000 BRICS
Initial Block Reward	50 BRICS
Halving Interval	Every 210,000 blocks (~4 years)
Transaction Fee	0.000005 BRICS (fixed)
Smallest Unit	0.00000001 BRICS
Target Block Time	10 minutes
Hashing Algorithm	SHA-256 (double-hash)

Table 4: Tokenomics Parameters

5.1 Emission Schedule

The block reward halves every 210,000 blocks:

```
reward(height) = 50 / 2 ^ floor(height / 210,000)
```

Halving	Block Range	Reward	Coins Minted	Cumulative
Genesis	0 - 209,999	50 BRICS	10,500,000	10,500,000
1st	210,000 - 419,999	25 BRICS	5,250,000	15,750,000
2nd	420,000 - 629,999	12.5 BRICS	2,625,000	18,375,000
3rd	630,000 - 839,999	6.25 BRICS	1,312,500	19,687,500
4th	840,000 - 1,049,999	3.125 BRICS	656,250	20,343,750

Table 5: Halving Schedule

6. Mining Protocol

6.1 Stratum v1

BricsCoin implements the Stratum v1 mining protocol on TCP port 3333, enabling direct compatibility with SHA-256 ASIC miners:

mining.subscribe -- Session initialization and extranonce assignment

mining.authorize -- Worker authentication (wallet_address.worker_name)

mining.set_difficulty -- Dynamic share difficulty adjustment

mining.notify -- Job distribution with block template and Merkle branches

mining.submit -- Share submission with nonce, timestamp, and extranonce2

6.2 BIP320 Version Rolling

The implementation supports BIP320 version rolling via **mining.configure**, allowing ASIC miners to use bits in the block version field as additional nonce space. The version rolling mask is **0x1FFFE000**, consistent with Bitcoin.

6.3 Hashrate Calculation

Network hashrate is estimated from accepted shares:

```
hashrate = (share_count x share_difficulty x 2^32) / time_window
```

7. Transaction Model

7.1 Legacy Transactions (ECDSA)

Transactions from "BRICS" addresses require a single ECDSA signature:

```
tx_data = sender || recipient || amount || timestamp
signature = ECDSA_sign(private_key, SHA-256(tx_data))
```

7.2 PQC Transactions (Hybrid)

Transactions from "BRICSPQ" addresses require both ECDSA and ML-DSA-65 signatures. The server performs three verification steps:

Hybrid signature verification: Both ECDSA and ML-DSA-65 must be valid

Address ownership: SHA-256(ecdsa_pubkey || dilithium_pubkey) must match sender

Balance check: Sender balance must cover amount + fee (0.000005 BRICS)

7.3 Migration Transactions

BricsCoin provides seamless migration from legacy ECDSA wallets to PQC hybrid wallets. The migration generates a new PQC wallet and transfers all funds in a single atomic transaction. Migration transactions are **fee-exempt** to incentivize adoption of quantum-safe addresses.

8. Network Architecture

8.1 API Layer

The RESTful API provides endpoints for:

Blockchain: Block retrieval, chain statistics, transaction history, explorer

Wallet: Balance queries, PQC wallet creation/recovery, transaction submission

Mining: Mining templates, active miner statistics, hashrate monitoring

Network: Peer management, node synchronization, network statistics

Security: PQC signature verification, security audit endpoints

8.2 Data Persistence

MongoDB serves as the persistent data store with indexed collections for blocks, transactions, wallets, mining statistics, and peer information.

9. Wallet Ecosystem

9.1 Web Wallet (PWA)

The browser-based wallet supports both legacy ECDSA and PQC hybrid wallets. All cryptographic operations run client-side. Wallet data is stored in localStorage and can be exported as encrypted JSON backups.

9.2 Desktop Wallet

Cross-platform Electron desktop wallet providing:

- Full PQC wallet creation with 12-word seed phrase
- Hybrid ECDSA + ML-DSA-65 transaction signing
- Wallet import and recovery from seed phrase
- Available for Windows (.exe), macOS (.dmg/.zip), and Linux (.AppImage)

9.3 Wallet Migration

A guided migration wizard enables users to transfer funds from legacy ECDSA wallets to new PQC hybrid wallets. The migration is fee-exempt and generates a complete PQC key set from a new seed phrase.

10. Security Analysis

10.1 Quantum Threat Model

Shor's algorithm, on a sufficiently powerful quantum computer, can break ECDSA by solving the elliptic curve discrete logarithm problem in polynomial time. BricsCoin's hybrid scheme provides defense-in-depth:

Scenario	ECDSA	ML-DSA-65	TX Valid?
Pre-quantum era	Secure	Secure	Yes
Quantum breaks ECDSA	Broken	Secure	No (ML-DSA required)
Both broken (hypothetical)	Broken	Broken	No
Classical attack only	Secure	Secure	Yes

Table 6: Quantum Threat Scenarios

10.2 Why Hybrid?

Defense in depth: If either algorithm has a vulnerability, the other provides a safety net.

Backward compatibility: Legacy wallets continue to operate with ECDSA-only signatures.

Proven + Novel: ECDSA has decades of validation; ML-DSA-65 provides quantum resistance.

NIST standardization: ML-DSA-65 was selected through a rigorous multi-year NIST evaluation and standardized as FIPS 204.

10.3 Client-Side Security

Private keys are generated and stored exclusively on the user's device. The server never receives or processes private keys. Only signatures and public keys are transmitted. For maximum security, the desktop wallet is recommended over the browser-based wallet.

11. Conclusion

BricsCoin represents a significant advancement in cryptocurrency security by implementing post-quantum cryptography at the consensus level. The hybrid ECDSA + ML-DSA-65 signature scheme, enforced for all PQC wallet transactions, ensures resilience against both classical and quantum computational attacks.

By combining Bitcoin's proven SHA-256 Proof-of-Work consensus and deflationary monetary policy with NIST-standardized post-quantum signatures, BricsCoin offers a practical path forward for cryptocurrency security in the quantum computing era.

BricsCoin is fully operational, open-source, and available for community participation.

Website: https://bricscoin26.org
Source Code: https://codeberg.org/Bricscoin_26/Bricscoin
Community: https://bricscoin26-chat.org/community
Twitter/X: @Bricscoin26

BricsCoin -- Created and developed by Jabo86
Copyright 2026 BricsCoin Project. All rights reserved.